

/*FTP server*/

#include<stdio.h>

#include<sys/types.h>

#include<netinet/in.h>

#include<string.h>

#include<stdlib.h>

#include<unistd.h>

int main()

{

FILE *fp;

int sd,newsd,ser,n,a,cli,pid,bd,port,clilen;

char name[100],fileread[100],fname[100],ch,file[100],rcv[100];

struct sockaddr_in servaddr,cliaddr;

printf("Enter the port address\n");

scanf("%d",&port);

sd=socket(AF_INET,SOCK_STREAM,0);

if(sd<0)

printf("Cant create\n");

else

printf("Socket is created\n");

servaddr.sin_family=AF_INET;

servaddr.sin_addr.s_addr=htonl(INADDR_ANY);

servaddr.sin_port=htons(port);

a=sizeof(servaddr);

bd=bind(sd,(struct sockaddr *)&servaddr,a);

if(bd<0)

printf("Cant bind\n");

else

printf("Binded\n");

listen(sd,5);

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clilen=sizeof(cliaddr);
newsd=accept(sd,(struct sockaddr *)&cliaddr,&clilen);
if(newsd<0)
{
printf("Cant accept\n");
}
else
printf("Accepted\n");
n=recv(newsd,rcv,100,0);
rcv[n]='\0';
fp=fopen(rcv,"r");
if(fp==NULL)
{
send(newsd,"error",5,0);
close(newsd);
}
else
{
while(fgets(fileread,sizeof(fileread),fp))
{
if(send(newsd,fileread,sizeof(fileread),0)<0)
{
printf("Can't send file contents\n");
}
sleep(1);
}
if(!fgets(fileread,sizeof(fileread),fp))
{
//when file pointer reaches end of file, file transfer "completed" message is send to accepted client
connection using newsd, socket file descriptor.
send(newsd,"completed",999999999,0);

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}
return(0);
}
}

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/*FTP Client*/

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#include<stdio.h>
#include<stdlib.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<unistd.h>

int main()
{
    FILE *fp;
    int csd,n,ser,s,cli,cport,newsd;
    char name[100],rcvmsg[100],rcvg[100],fname[100];
    struct sockaddr_in servaddr;
    printf("Enter the port");
    scanf("%d",&cport);
    csd=socket(AF_INET,SOCK_STREAM,0);
    if(csd<0)
    {
        printf("Error....\n");
        exit(0);
    }
    else
    printf("Socket is created\n");
    servaddr.sin_family=AF_INET;
    servaddr.sin_addr.s_addr=htonl(INADDR_ANY);
    servaddr.sin_port=htons(cport);
    if(connect(csd,(struct sockaddr *)&servaddr,sizeof(servaddr))<0)

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printf("Error in connection\n");
else
printf("connected\n");
printf("Enter the existing file name\t");
scanf("%s",name);
printf("Enter the new file name\t");
scanf("%s",fname);
fp=fopen(fname,"w");
send(csd,name,sizeof(name),0);
while(1)
{
s=recv(csd,rcvg,100,0);
rcvg[s]='\0';
if(strcmp(rcvg,"error")==0)
printf("File is not available\n");
if(strcmp(rcvg,"completed")==0)
{
printf("File is transferred.....\n");
fclose(fp);
close(csd);
break;
}
else
fputs(rcvg,stdout);
fprintf(fp,"%s",rcvg);
return 0;
}
}

```